

Collective Seed Storage

A Sustainable Approach to Ex Situ Seed Conservation through Citizen Participation and Distributed Storage

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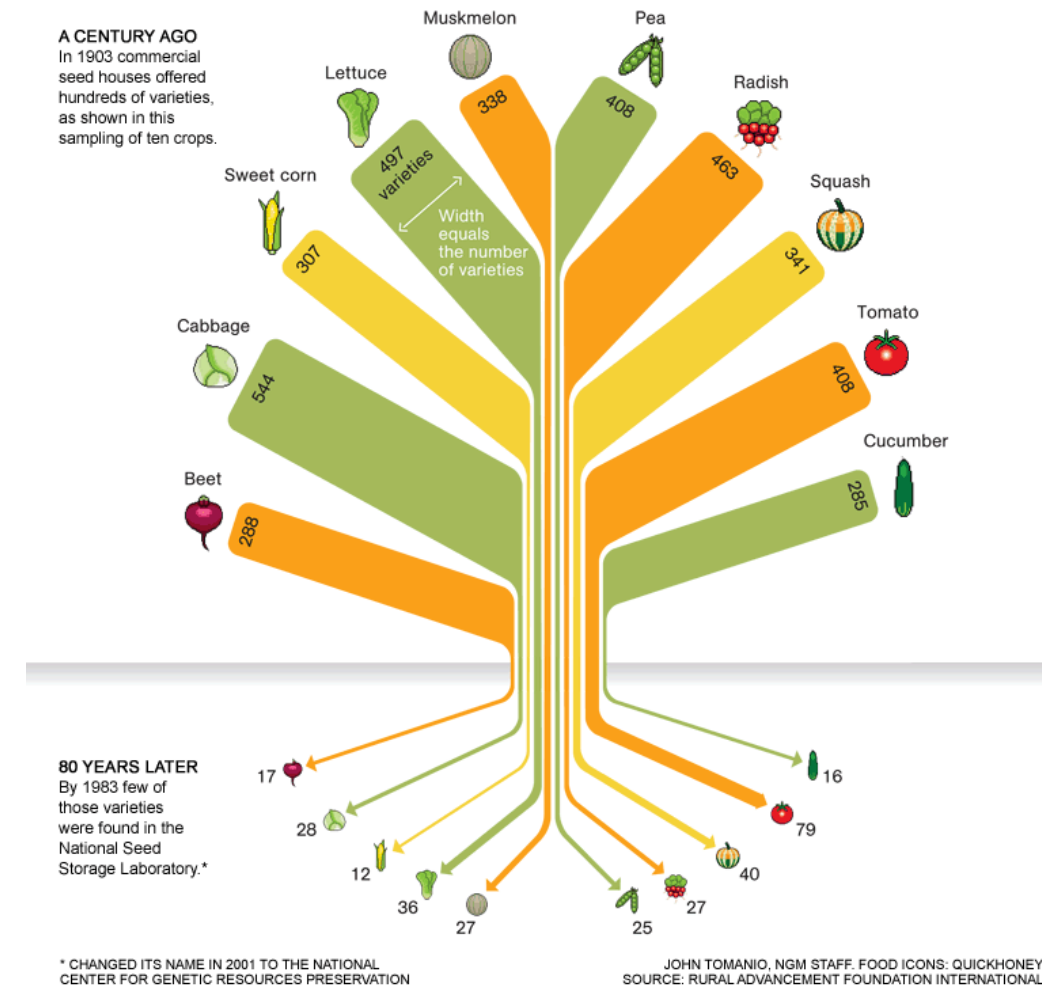


The Need for Seed Diversity

- Over 30,000 edible plants exist worldwide, but just 15 provide 90% of food energy intake.
- More than 95% of crop **genetic erosion** studies report changes in diversity, with nearly 80% showing evidence of loss [1].
- Crop Wild Relatives (CWRs) hold genetic resources absent from cultivated varieties, useful to improve productivity, resilience and quality.

Genetic diversity

Vital to adapt to climate change, pests, diseases and evolving societal needs



Conservation Strategies

Criteria	In Situ	Ex Situ
Location	Natural environment / farms	Seed banks, seed vaults, labs
Evolution	Plants adapt naturally	Static; no natural evolution
Risk of Loss	Higher (land-use, climate)	Lower (controlled conditions)
Cultural Link	Strong	Weak or absent
Access & Sharing	Local and informal	Global, formal (e.g. SMTA)
Best Use	Wild relatives, landraces	Major crops, backup, breeding

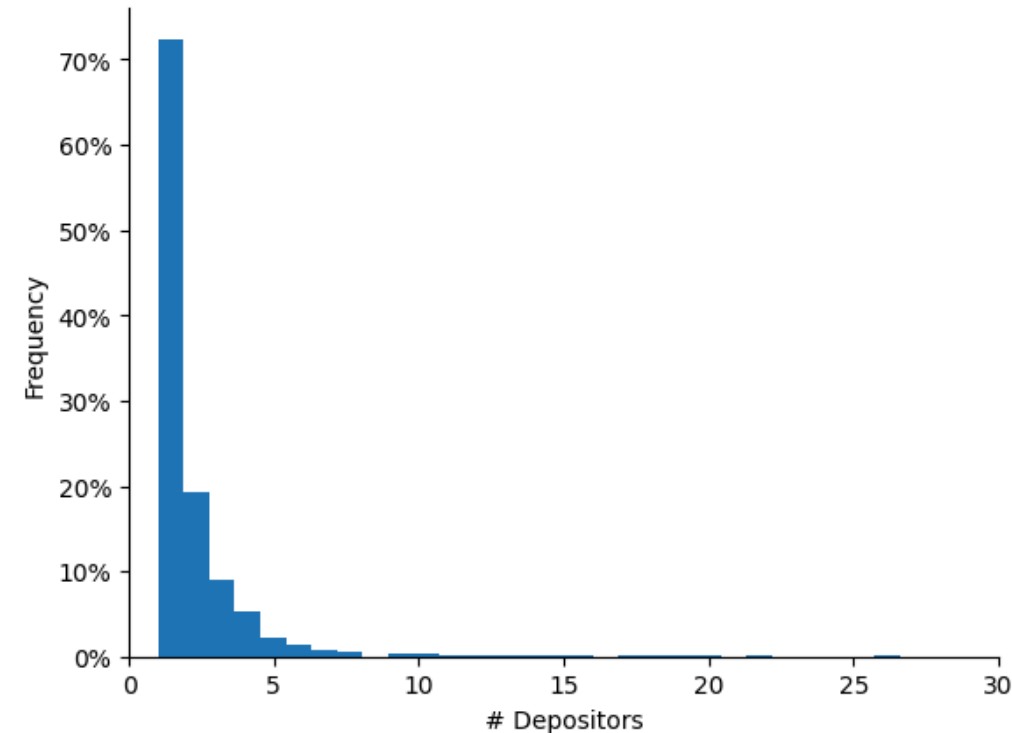
Seed Banks and the SGSV

- A seed bank safely conserves the genes that make each plant variety unique, and keeps them available for future use.
- The **Svalbard Global Seed Vault (SGSV)** is unanimously considered the global **backup** facility for seed banks worldwide.
- 2023: **1,267,127 samples**, from **>100 genebanks**, representing **>6,000 species**.
- Only depositors can withdraw their own seeds (unlike the Millennium Seed Bank).



Open Problem

- Our analysis of the SGSV **Seed Portal** shows that the vast majority of species have a **single depositor**.
- *Oryza sativa* (Asian rice): **29 depositors**, 171,193 accessions, 136 countries.
- *Hygroryza aristata* (a CWR of rice): **1 depositor**, 4 accessions, 1 country.
- **The Threat:** if a disaster hits both the primary seed bank and the vault, low-redundancy species could be lost forever.



Our Vision: Collective Seed Storage (CSS)

- **Key insight:** the temperature conditions of domestic freezers can meet FAO standards for long-term seed conservation.
- **Shared-economy approach:** repurpose a tiny, idle portion of home freezers to store seeds.
- **First attempt** at a peer-to-peer infrastructure to preserve seed diversity, complementing centralized genebanks.
- Each household becomes a micro-node, increasing redundancy with minimal infrastructure investment.



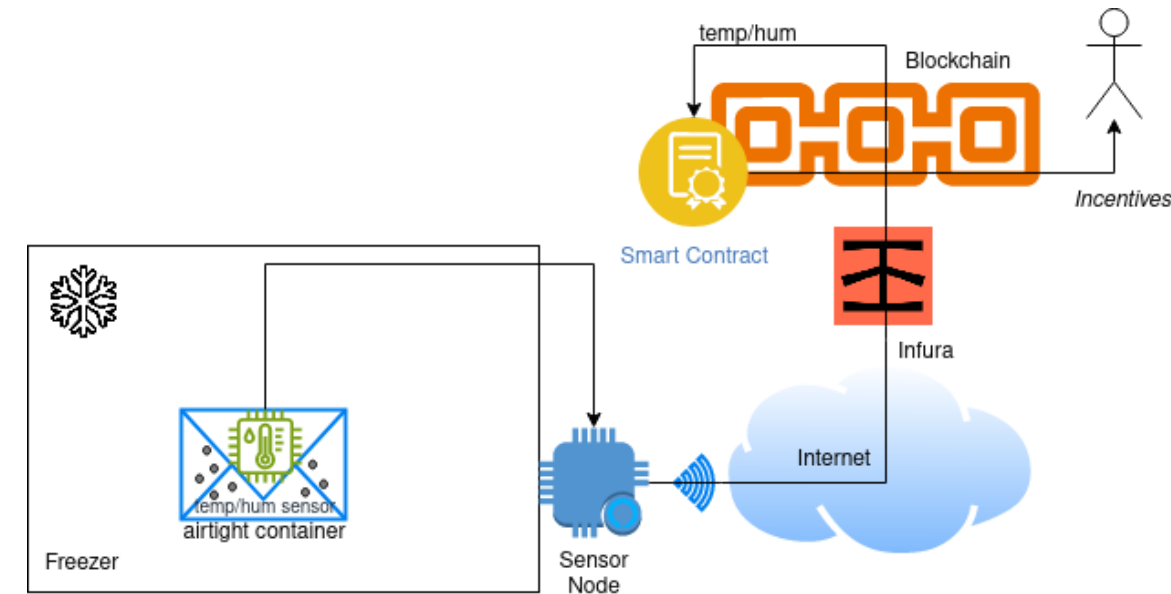
CSS Requirements

To ensure feasibility and active participation, the system must meet:

1. **Storage Standards (R1):** comply with FAO long-term storage standards (approx. -18°C). Subzero acceptable if ideal conditions unavailable.
2. **User-Friendly (R2):** simple installation with off-the-shelf equipment; only home WiFi assumed.
3. **Longevity (R3):** run for at least 1 year without user intervention.
4. **Incentivization (R4):** support suitable incentive mechanisms to reward long-term participation.

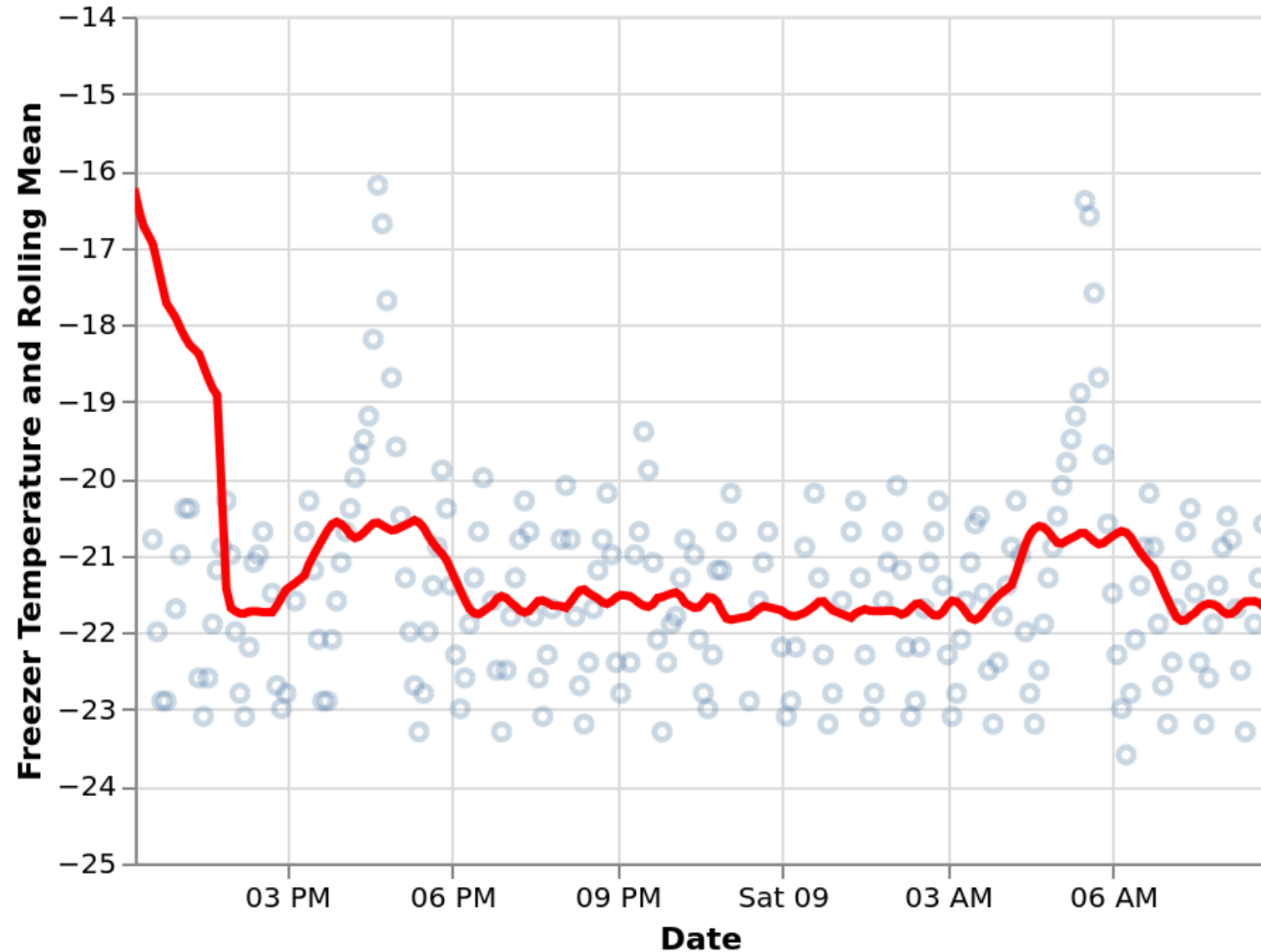
Proof-of-Concept: Architecture

- **Sensor:** DHT22 (temperature/humidity) inside an airtight seed container in the freezer.
- **Sensor Node:** ESP32 Devkit V1 placed *outside* the freezer, connected via three tiny wires (R2).
 - Placing the ESP32 inside hinders WiFi (>40% packet loss) and battery efficiency drops in the cold.
- **Communication:** ESP32 transmits readings via WiFi to a Blockchain smart contract that dispenses incentives (R4).



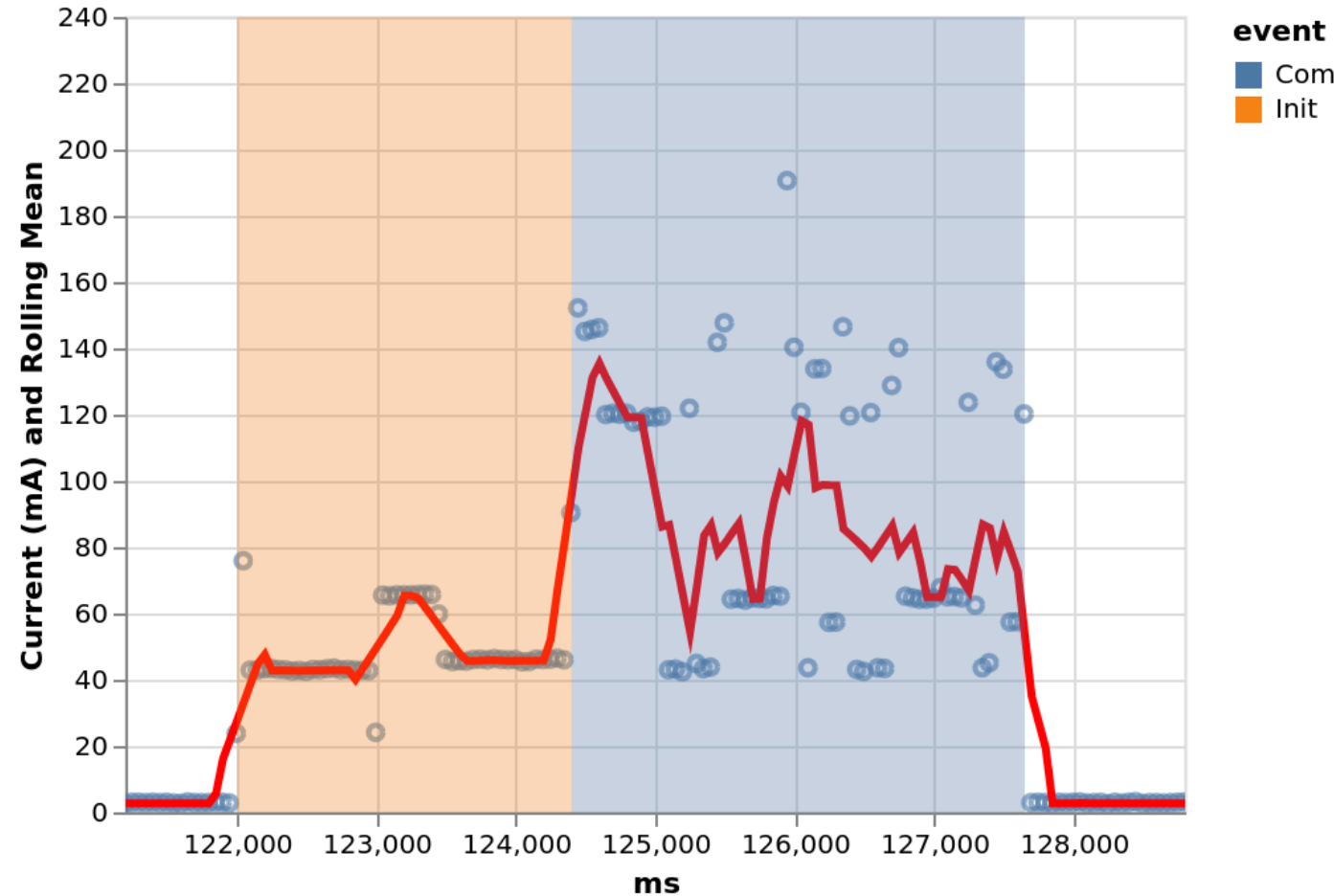
R1: Storage Conditions

- Monitored a **BOSCH KGN39X23** domestic freezer set to -18°C , samples every 5 minutes.
- After an initial transient, temperature **oscillates** noticeably.
- Ideal Standard 4.2.3 not fully met, but conditions are consistent with **acceptable subzero storage**.



R3: Energy Consumption &

- **init task:** ~2400ms @ ~50mA ($\approx 0.03\text{mAh}$)
- **com task:** ~3160ms @ ~100mA ($\approx 0.09\text{mAh}$)
- **deep sleep:** 10 μA nominal
- 3.6V, 1.5Ah LiFePo4 battery
- With a sleep period $X \geq 0.73$ hours, the node meets the **1-year lifetime** requirement.



R4: Incentives

- Blockchain-based **lottery**, inspired by DePIN (token-incentivized participatory sensing).
- Users are eligible to enter **only if** their readings stay within an acceptable storage range.
- Minimalistic smart contract deployed on Sepolia testnet, accessed via the Infura API.
- Up to 32 packed readings per transaction, reducing communication overhead.



CSS contributes to several UN Sustainable Development Goals

- **SDG 2 — Zero Hunger:** distributed backup infrastructure against genetic erosion
- **SDG 12 — Responsible Consumption & Production:** reuses existing freezer capacity instead of new infrastructure
- **SDG 13 — Climate Action:** preserves genetic resources needed for climate adaptation
- **SDG 17 — Partnerships for the Goals:** potential integration with the Multilateral System (MLS)

Future Work

- **Long-term storage studies:** how close can domestic freezers get to ideal Standard 4.2.3 conditions? Variability across freezer models.
- **Energy harvesting:** exploit the inside/outside thermal gradient for energy-neutral operation.
- **User studies:** willingness to participate; game-theoretic engagement mechanisms.
- **Institutional integration:** connect CSS with the Multilateral System (MLS) and governance of seed exchanges.
- Evaluate economic costs and long-term participation strategies.

Conclusions

- Even the SGSV, the world's most renowned backup facility, shows **limited redundancy** for many species.
- **CSS** is a shared-economy vision: domestic freezers as a distributed infrastructure for long-term seed storage.
- Our Proof-of-Concept demonstrates the **technical feasibility** of IoT-based monitoring and blockchain-based incentives.
- Acceptable subzero storage conditions are achievable in ordinary domestic freezers.



Thank you!

Questions?

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 <https://andreavitaletti.github.io/>

<https://github.com/andreavitaletti/slides/blob/main/ICAISF2026/presentation.pdf>

Code:

https://github.com/andreavitaletti/PlatformIO/tree/main/Projects/web3E_SC



Image sources

- Slide 1, 6, 12: generated by Gemini
- Slide 2: <https://medium.com/thenextnorm/importance-of-genetic-diversity-in-agriculture-b9f88f5fda55>
- Slide 4:
https://en.wikipedia.org/wiki/Svalbard_Global_Seed_Vault#/media/File:Svalbard_Global_Seed_Vault_February_2025.jpg
- Slide 5: <https://doi.org/10.1111/rec.13174>